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Lightweight Fall Detection Based on FMCW Micro-Doppler and Power Burst Signatures Extraction

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***Abstract*—Nowadays, the aging problem is intensified, and the demand for the elderly fall detection at home is increased. The fall detection system based on millimeter-wave frequency modulated continuous wave (FMCW) radar shows its advantages. This paper proposes a lightweight fall detection system based on FMCW radar. The noise is removed by ordered statistics constant false alarm rate (OS-CFAR) filter. After that, the power burst curve (PBC) is used to detect motion events. The micro-Doppler frequency estimations of human body parts are extracted by CLEAN, density-based spatial clustering of applications with noise (DBSCAN) and k-nearest neighbors (KNN) algorithm. Finally, 1D convolutional neural network (1D-CNN) and bidirectional long short-term memory (Bi-LSTM) network are used to make the final detection. The accuracy of the fall detection model is 93.23%, with precision, recall and F1 score exceeding 90%.**

Keywords—Deep learning, fall detection, lightweight network, millimetre-wave radar, radar signal processing.

I. Introduction

According to the 2024 population statistics, the world population aged over 65 is approximately 1.29 times that of the population aged under 5 years [1]. In the case of limited medical staff, a monitoring system that can operate independently in the elderly's home is becoming more and more important. The monitoring system is required to achieve low missed detection rate, low false alarm rate, whole-process monitoring, and user friendliness [2].

Currently, there are three commonly used methods to extract human gait features: wearable sensors, optical sensors and radar sensors [3]. Regarding the application of millimeter-wave frequency modulated continuous wave (FMCW) radar to extract human gait features, Yao et al. [4] use multilevel thresholding OTSU method to binarize the velocity-time spectrum, and use image mask to remove noise, Then the feature envelopes of the velocity-time spectrum are extracted according to the energy percentage. Yao et al. [5] use short time Fourier transform (STFT) to obtain range-velocity gait features from radar data, and propose to detect motion events of human gait features with a sliding window method.

For the recognition of human behavior, machine learning is used in the past. Liu et al. [6] use support vector machine (SVM) and k-nearest neighbor (KNN) algorithm to classify human behavior. Now, a large number of human behavior recognition methods are implemented by deep learning. Wang et al. [7] propose a line kernel convolutional neural network (LKCNN), which directly processes baseband data to detect fall events. Hsu et al. [8] use bidirectional long short-term memory (Bi-LSTM) network to complete fall detection based on range-doppler histogram.

Previous fall detection models often use micro-Doppler spectrogram for direct training, which leads to a large amount of computation. In this paper, the motion events are detected, and the one-dimensional micro-Doppler frequency estimations of different parts are extracted from the velocity-time spectrum by using power burst curve (PBC). Finally, a lightweight network is proposed, which uses one-dimensional curves to detect falls.

The rest of this paper is organized as follows. The Section II introduces the FMCW radar, radar parameters setting and data acquisition method. The Section III introduces the signal model and signal processing methods, including radar data preprocessing and feature extraction. The Section IV introduces the lightweight deep learning network architecture for fall detection, and gives the analysis and discussion of the results. Finally, we conclude this paper in Section V.

II. Experimental Setup and Measurement

The IWR1443 and DCA1000 kit of Texas Instruments (TI) are used to collect human gait data. Among them, IWR1443 is a 76-81GHz millimeter wave sensor with 3 transmit (TX) antennas and 4 receive (RX) antennas. We use 1 TX antenna and 4 RX antennas, and set the starting frequency to 77GHz. The DCA1000 transmits large amounts of radar data in real time to the PC for storage and further processing. The detailed radar parameters are shown in Table 1.

The experiment is conducted in a classroom, as shown in Figure 1, where the radar height is 1.5m, the subjects' activity area is a 10m × 2m rectangle, 1.3m away from the radar. The designed motion patterns and corresponding labels are shown in Table 2. In order to ensure the safety of subjects, subjects are required to fall on the cushion. Subjects include 6 males and 2 females, ranging in age from 21 to 24, with heights ranging from 1.63m to 1.83m, and weights ranging from 51kg to 93kg. Each motion pattern is repeated 10 times, and 960 samples of radar data are collected.

III. Data Preprocessing and Feature Extraction

A. Signal Model and Preprocessing

The signal of RX and the signal of TX are mixed to get the Intermediate frequency (IF) signal. IF signal passes through Analog-to-Digital Converter (ADC) to obtain raw radar data. Average background subtraction is adopted to eliminate the noise effects generated by other static objects in the experimental environment [9]. Then, STFT is used to obtain the range-

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velocity-channel data cube of each frame. Finally, the range-velocity-time data cube is obtained by incoherent superposition along the receive channel dimension.

TABLE 1. PARAMETERS FOR EXPERIMENTS

Parameters	Quantity
Starting frequency	77GHz
Frequency slope	39.058MHz/μs
Bandwidth	1999.77MHz
Ramp end time	51.2μs
Idle time	61.2μs
Frame	30ms
ADC samples	256
Number of frames	200
Number of chirps per frame	200
Number of TX/RX	1/4
Sample rate	10000Ksps

TABLE 2. DESIGNED MOTION PATTERNS

	Motion Patterns	Labels
1	Fall while walking toward the radar	Fall
2	Fall while walking away from the radar	Fall
3	Fall toward the radar while sitting down	Fall
4	Fall backward the radar while sitting down	Fall
5	Fall toward the radar while bending to pick something	Fall
6	Fall backward the radar while bending to pick something	Fall
7	Walk toward the radar	Non fall
8	Walk away from the radar	Non fall
9	Sit down while walking toward the radar	Non fall
10	Sit down while walking away from the radar	Non fall
11	Bend to pick something while walking toward the radar	Non fall
12	Bend to pick something while walking away from the radar	Non fall

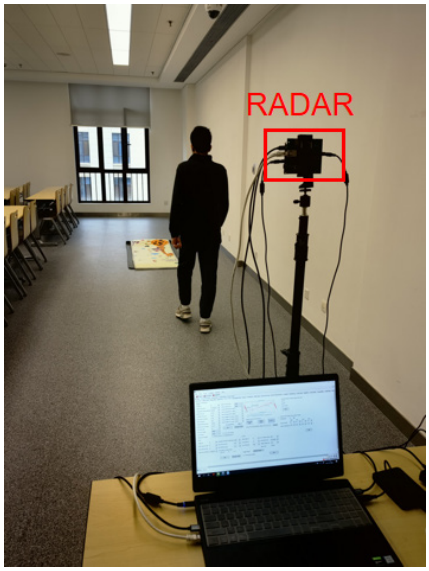


Figure 1. Experimental environment.

B. Feature Extraction

Figure 2 shows the processing flow of feature extraction. The block diagram begins with data cube obtained from radar signal preprocessing. Order-statistic constant false alarm rate (OS-CFAR) filter is applied to identify human radar signal, then motion events are detected by PBC. The signals of different parts of human body are extracted, and the micro-Doppler frequency estimations of body parts are fitted at last. Details of the different processing steps are described as follows.

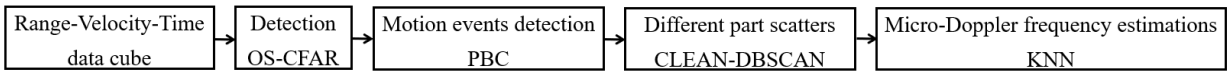


Figure 2 The processing flow of feature extraction.

OS-CFAR detection: In order to obtain human radar signals in a noisy environment, OS-CFAR is applied to the range-velocity spectrum corresponding to each frame of the range-velocity-time data cube obtained above [10]. The reference window cells are centered on the test cell, with training cells forming a 31 × 31 rectangular window and guard cells comprising an inner 15 × 15 rectangular window. For the range-velocity spectrum corresponding to each motion pattern, the farthest distance from the radar during the subjects' movement is cut in the distance dimension to eliminate multi-path echo interference. In the velocity dimension, data are cut from -0.7m/s to 2.5m/s for movement toward the radar, and from -2.5m/s to 0.7m/s for movement away from the radar to reduce the computational load. After OS-CFAR processing, the range-velocity-time data cube of human gait is obtained. In order to reduce the influence of scattering intensity of velocity-time spectrum caused by different distances, the range-velocity spectrum of each frame is normalized, and the maximum power of each frame is selected for dimensionality reduction to obtain the velocity-time spectrum [11]. The velocity-time spectrum of falling while walking away from radar obtained by OS-CFAR is shown in Figure 3(a).

PBC extraction: It can be seen from Figure 3(a) that the subject remains stationary after falling. In order to reduce the amount of subsequent processing computation data, PBC is used to detect motion event from velocity-time spectrum [12]. The PBC is obtained by calculating the sum of normalized power within each frame of the velocity-time spectrum, In order to obtain the threshold of PBC, a moving average filter with a length of 0.3s is used to filter the PBC and obtain the filtered PBC, and the threshold is obtained by the following equation [13]

$$T = S_{f_{min}} + \alpha \times \left(S_{f_{max}} - S_{f_{min}}\right) \quad (1)$$

where $S_f(n)$ is the filtered PBC, $S_{f_{\min}}$ and $S_{f_{\max}}$ are the minimum and maximum values of $S_f(n)$ respectively, α is the empirical coefficient, it is set to 0.15 in algorithm implementation. The PBC, the filtered PBC and the threshold are shown in Figure 3(b).

Scatters extraction: In order to reduce the computational load of the subsequent deep learning model, the micro-Doppler frequency estimations of torso, arms and legs are extracted from

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the two-dimensional velocity-time spectrum. Figure 4(a) shows the velocity-time spectrum of falling while walking away from radar after motion event detection. Combined CLEAN and clustering algorithms, the corresponding scatters of torso, arms and legs are obtained [14]. The scatters of different parts of human body are extracted by CLEAN algorithm. The embedded DBSCAN makes further judgment on the different parts of the scatters obtained from CLEAN according to the scatter density, thus improving the accuracy of scatter extraction.

The micro-Doppler frequency estimations: In order to further separate the corresponding left and right arm and leg scatters from the arm and leg scatters, it is necessary to find out the periodicity of the movement adaptively. One of the characteristics of human gait is the maximum speed of the human torso when the feet touch the ground [15]. Using the velocity-time spectrum obtained after OS-CFAR processing, the maximum power at each frame is extracted to obtain the maximum power velocity-time curve, and the peak interval of it is the motion period. The corresponding left and right arm and leg scatters can be separated according to the motion period. Finally, KNN algorithm is used for regression fitting to obtain the micro-Doppler frequency estimations corresponding to different human body parts [16]. K value is set to 6 and Euclidean distance is adopted. Five micro-Doppler frequency estimations can be obtained for each sample. The scatters and micro-Doppler frequency estimations obtained from the velocity-time spectrum of falling while walking away from radar through the above processing are shown in Figure 4(b).

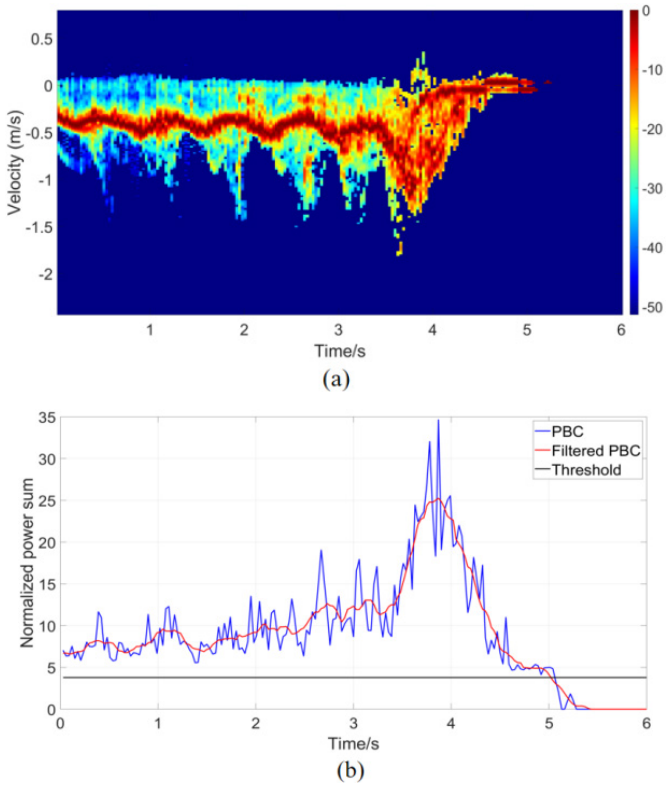


Figure 3. Falling during walking away from radar: (a) the velocity-time spectrum, (b) the PBCs and the threshold line.

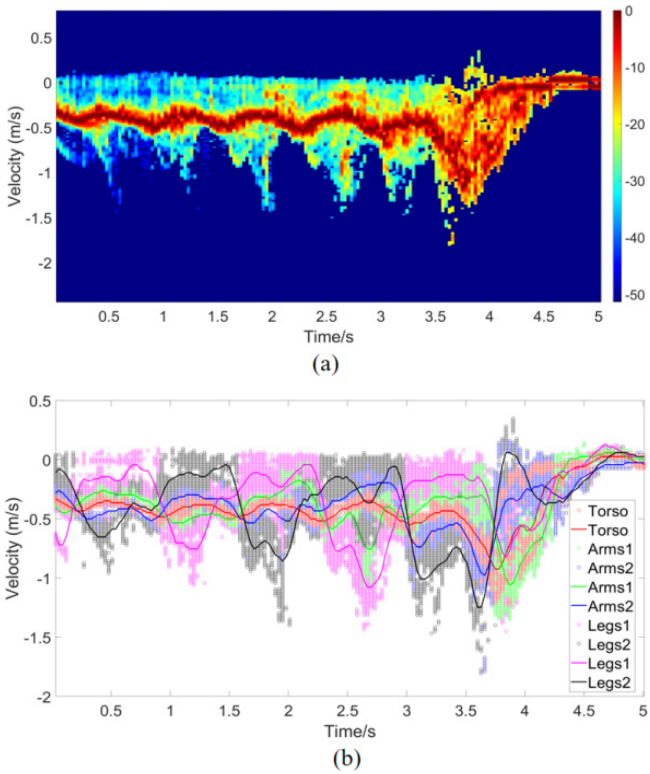


Figure 4. Falling during walking away from radar: (a) Motion event detection of Figure 3(a), (b) scatters and the micro-Doppler frequency estimations of Figure 4(a).

IV. Deep Learning Model and Result

A. Fall Detection Model

Bi-LSTM is more efficient than ordinary recurrent neural networks (RNN). After signal processing in Section III, five micro-Doppler frequency estimations of different human body parts and 1 PBC can be obtained from each sample. These curves are one-dimensional time series and can be used as the input of Bi-LSTM. We propose a lightweight fall detection model based on 1D convolutional neural network (1D-CNN) and Bi-LSTM.

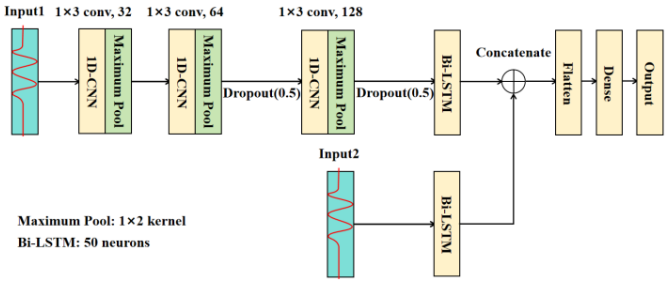


Figure 5. The architecture of fall detection model

The architecture of fall detection model proposed is shown in Figure 5, where input 1 are micro-Doppler frequency estimations of body parts and input 2 are power burst curves. The data feature of the micro-Doppler frequency estimations of body parts is extracted by 1D-CNN layers, and the maximum pooling layers are used to reduce the dimension of the data feature. The convolution kernel sizes of the three 1D-CNN layers are all 1×3, the channel numbers are 32, 64 and 128. The

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kernel sizes of the three maximum pooling layers are all 1×2, and the stride of both the convolution layer and the pooling layer are all 1. Finally, the Bi-LSTM layer is used for fall detection. In addition, PBC performs fall detection through the Bi-LSTM layer, and the number of neurons in both Bi-LSTM layers is 50, the detection results of the two inputs are fused in time dimension, and the final detection results are obtained through the full connection layer. Regularization terms with coefficient 0.5 are introduced into the network to prevent overfitting during model training.

B. Results and Analysis

The first 6 subjects' micro-Doppler frequency estimations of body parts and PBCs are used to train, and the last 2 subjects' micro-Doppler frequency estimations of body parts and PBCs are used to test. In order to evaluate the model, accuracy, precision, recall and F1 score are used as evaluation indicators of the model, and the four evaluation indicators of the test result are shown in Table 3. As can be seen from Table 3, the accuracy of model is 93.75%, with precision, recall and F1 score exceeding 92%. In addition, the model adopts one-dimensional features for training, which reduces the training time consumption compared with the two-dimensional micro-Doppler signature model.

Considering the gait differences between different subjects, *k*-fold cross validation is used to investigate the generalization of the model [17]. Seven subjects for training and 1 subject for testing. The whole process is repeated for 8 times, and the average of evaluation indicators of the 8 test results is taken. The corresponding results of *k*-fold cross validation are shown in Table 4. The accuracy of *k*-fold cross validation is 93.23%, with precision, recall and F1 score exceeding 90%. This shows the method proposed has good generalization.

TABLE 3. FOUR EVALUATION INDICATORS OF THE TEST RESULT

Accuracy	Precision	Recall	F1 score
93.75%	95.18%	92.17%	93.65%

TABLE 4. RESULTS OF K-FOLD CROSS VALIDATION

Accuracy	Precision	Recall	F1 score
93.23%	96.21%	90.21%	92.82%

TABLE 5. COMPARISON WITH RELATED WORK

Work	Number of patterns		Number of samples		Evaluation
	Fall	Non fall	Fall	Non fall	
[18]	4	3	408	117	Accuracy: 87%
[19]	3	3	600	400	Accuracy: 96.6% Precision: 95.2%
Ours	6	6	480	480	Accuracy: 93.23% Precision: 96.21% Recall: 90.21% F1 score: 92.82%

Finally, Table 5 compares the method proposed in this paper with related work, including number of patterns, number of samples and model evaluation indicators. It can be seen from Table 5, under the premise of reducing the amount of computation, the accuracy of the proposed method is not significantly reduced, and the precision is slightly improved.

V. CONCLUSION

A lightweight fall detection system based on FMCW radar is proposed in the paper. Noise is removed by OS-CFAR filter, The PBC threshold is used to detect motion events, and micro-Doppler frequency estimations of human torso, arms and legs are extracted from the velocity-time spectrum of human gait. Finally, a lightweight fall detection model based on 1D-CNN layers and Bi-LSTM layer is constructed, and the classification results of micro-Doppler frequency estimations and PBC are fused to realize fall detection. The accuracy of model is 93.75%, with precision, recall and F1 score exceeding 92%. In addition, *k*-fold cross validation is used to investigate the generalization of the model. The accuracy is 93.23%, with precision, recall and F1 score all exceeding 90%.

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